Yassa Ashraf

Software Engineer | Backend, Cloud & AI Systems

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PROFESSIONAL SUMMARY

Software Engineer specializing in backend, cloud, and AI-enabled systems across enterprise and product environments. Experienced in TypeScript, Node.js, NestJS, Java, Spring Boot, ASP.NET, React, Next.js, and AWS, with hands-on delivery across APIs, microservices, event-driven systems, infrastructure as code, CI/CD, and production operations. Strong background in scalable system design, cloud-native development, application reliability, and integrating AI capabilities into real-world software products.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Java, Python, C#, SQL, Dart, Bash
Backend: Node.js, NestJS, Express, Spring Boot, ASP.NET Web API, REST, GraphQL, Microservices
Frontend & Mobile: React, Next.js, Angular, React Native, Flutter, Tailwind CSS
Cloud & DevOps: AWS, EC2, Lambda, ECS, EKS, S3, API Gateway, CloudFront, CloudWatch, SageMaker, Terraform, Docker, Kubernetes, GitHub Actions, Jenkins, CI/CD
Data: PostgreSQL, MySQL, MongoDB, DynamoDB, Redis, Prisma, Mongoose, JDBC
AI & ML: Generative AI, LLMs, RAG, Agentic AI, Multi-Agent Systems, Hugging Face Transformers, ChemBERTa, Gemini, Google ADK, Embeddings, Semantic Retrieval, Explainable AI
Architecture & Security: Clean Architecture, Domain-Driven Design, Event-Driven Architecture, Serverless Architecture, Distributed Systems, OAuth2, JWT, RBAC, API Security
Testing & Observability: JUnit, Jest, Unit Testing, Integration Testing, CloudWatch, Grafana, Logging, Monitoring

PROFESSIONAL EXPERIENCE

Mid-Level Software Engineer — Deloitte Cairo, Egypt (Hybrid)
Sep 2026 – Present | Full-time

- Design and build cloud-native backend services and platform components using AWS, TypeScript, Node.js, Terraform, and modern software engineering practices.
- Develop secure APIs, event-driven workflows, and infrastructure components for enterprise platforms operating in regulated environments.
- Implement infrastructure as code and CI/CD workflows supporting repeatable validation, deployment, and environment consistency.
- Collaborate with cross-functional and international teams on cloud architecture, integration, testing, security, and production delivery.

Senior Software Engineer — EV Cozmo Greater London, UK (Remote)
Jul 2026 – Sep 2026 | Contract

- Reverse-engineered mobile clients, local data stores, and network behavior across 10+ EV charging applications to integrate station and live-availability data where public APIs were incomplete or unavailable.
- Recovered and normalized station inventories from Firestore caches, local application storage, and network responses, significantly increasing charging-location coverage across integrated networks.
- Architected a pluggable live-status provider layer synchronizing multiple charging networks in parallel with source-level timeouts, rate-limit-aware authentication, and fail-soft error handling.
- Built geospatial entity-resolution and deduplication pipelines using MongoDB geospatial queries and GeoJSON indexing to map third-party charging records to canonical stations.
- Developed clean-architecture station services for CRUD, nearest-station search, favorites, and image management using AWS Lambda, API Gateway, and Mongoose.
- Delivered Flutter station-discovery features including maps, filters, favorites, and production releases to the Apple App Store and Google Play.
- Maintained CI/CD pipelines with GitHub Actions and SST across development and production environments.

Associate Software Engineer — Vodafone Intelligent Solutions (VOIS) Cairo, Egypt
Jul 2025 – Sep 2026 | Full-time

- Developed and enhanced enterprise ASP.NET Web API services and Angular applications using reusable, maintainable architecture patterns.
- Implemented advanced filtering, pagination, validation, and shared backend components across multiple enterprise applications.
- Refactored frontend and backend modules to reduce duplicated logic and improve maintainability.

- Strengthened application reliability through improved error handling, logging, validation, and service-layer design.
- Supported UAT, stakeholder demonstrations, release preparation, and production-oriented delivery under enterprise security requirements.

Software Engineer Intern — Vodafone Intelligent Solutions (VOIS) Cairo, Egypt
Feb 2025 – Jul 2025 | Internship

- Contributed to ASP.NET Web API backend enhancements and Angular frontend development.
- Improved performance of critical endpoints through database query and API optimization.
- Implemented unit tests and participated in peer reviews and software quality processes.

AI Software Engineer — VIXI AI Remote, UAE
Sep 2025 – Jan 2026 | Contract

- Designed NLP and retrieval pipelines using transformer-based models for enterprise AI use cases.
- Built containerized AI inference services with deployment automation, autoscaling, and monitoring.
- Collaborated with product and data teams to translate business requirements into production AI capabilities.
- Improved development velocity by introducing preview and staging environments for faster iteration.

Software Engineer — Xylem Software Remote, Egypt
Jun 2025 – Nov 2025 | Contract

- Delivered web and mobile functionality in a monorepo using React, React Native, Node.js, and shared application services.
- Led backend development for analytics and operational platforms, including ingestion APIs and dashboard services.
- Improved application performance through caching, query optimization, and backend service improvements.

Mid-Level Backend Engineer — Trevi Cairo, Egypt
Oct 2024 – Sep 2025 | Full-time

- Architected wallet and loyalty backend services using NestJS, Prisma, and relational databases.
- Integrated cashback and promotion workflows supporting transaction-heavy financial use cases.
- Deployed services on AWS with health checks, retry mechanisms, monitoring, and reliability controls.

Software Engineer — MM Solutions Remote, USA
Jul 2024 – Mar 2025 | Part-time

- Led web integration work and optimized frontend loading, caching, and application performance.
- Automated dispatch and operational workflows supporting Amazon Freight partner operations.

Full Stack Developer — Kinetics Egypt Cairo, Egypt
Aug 2024 – Oct 2024

- Delivered cloud-hosted software solutions using AWS EC2 and S3 with secure environment configuration.
- Developed product features in collaboration with engineering and product teams under tight delivery timelines.

Full Stack Developer — Fiverr Remote
May 2023 – Oct 2023 | Freelance

- Built custom web applications, dashboards, and interactive products for international clients.
- Managed delivery from requirements gathering and implementation through deployment and handover.

Full Stack Developer Intern — Instatus Cairo, Egypt
Jul 2022 – Sep 2022 | Internship

- Developed frontend functionality using Next.js, GraphQL, and Tailwind CSS.
- Integrated GraphQL queries and mutations for real-time status-management workflows.
- Improved UI responsiveness and collaborated with backend engineers on schema integration.

SELECTED PROJECTS

Beity App — Cloud Kitchens Ordering and Delivery System NestJS, React Native, AWS, PostgreSQL

- Built a cloud-kitchen platform supporting ordering, delivery management, users, and operational workflows.
- Implemented order lifecycle management, role-based authorization, clean backend architecture, and AWS deployment pipelines.

Intervyo — Smart Interview Platform Next.js, Node.js, TypeScript, PostgreSQL

- Developed an interview-management platform supporting technical and HR workflows, scheduling, candidate evaluation, and REST APIs.

E-Commerce Platform for Plastic Pallets NestJS, Next.js, Kafka

- Designed product, order, and payment microservices using Kafka for asynchronous communication and resilient service integration.

Bookstore Serverless Application AWS Lambda, API Gateway, DynamoDB

- Built serverless CRUD and search APIs with authentication, rate limiting, DynamoDB access patterns, CloudWatch monitoring, and CI/CD.

ECS Help Desk System MERN Stack, Tailwind CSS

- Built a ticket-management platform supporting roles, SLAs, real-time updates, analytics, and operational workflows.

EDUCATION

Bachelor of Computer Science — Software Engineering Oct 2021 – Jul 2025
German International University, Cairo

RESEARCH & PUBLICATIONS

Bachelor Thesis — Using Explainable AI Algorithms for Material Recognition Grade: A+

- Built ChemBERTa-based material-recognition models with post-hoc explainability techniques.
- Enabled traceable predictions using attention visualization and feature-attribution methods.
- Deployed Flask inference services using AWS SageMaker, EC2, and S3 with CI/CD pipelines.
- Delivered a Next.js dashboard for real-time inspection of model predictions and explanations.

MoleculeX-Predictive: An augmentation of transformers with explainable AI and knowledge layers for efficient drug property prediction
Yassa Ashraf, Daniel Sameh, John Zaki, Amal Aboulhassan
Array (Elsevier), 2026
DOI: 10.1016/j.array.2026.100820 | ScienceDirect

CERTIFICATIONS

Oracle Cloud Infrastructure Foundations Associate (2025) Issued Mar 2025
Credential Verification
Career Essentials in Generative AI — Microsoft / LinkedIn Issued Oct 2025
Explainable AI (XAI) — Duke University Issued Oct 2025
Interpretable Machine Learning — Duke University Issued Oct 2025
Ethics in the Age of Generative AI — LinkedIn Issued Oct 2025

LANGUAGES

English: Professional Proficiency | **Arabic:** Native Proficiency | **German:** Limited Working Proficiency